The Internet of Things (IoT)

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Introduction to the Internet of Things

The Internet of Things (IoT) is a network of physical objects, such as appliances, vehicles, machines, wearables and sensors, that are embedded with sensors, software and connectivity so they can collect data and exchange it with other devices and systems over the internet. In simple words, IoT means everyday things that are connected to the internet and can sense, communicate and sometimes act on their own. The term "Internet of Things" was coined by the British technologist Kevin Ashton in 1999 while he was working at Procter and Gamble and later co-founded the Auto-ID Center at MIT. Analysts estimate that there are now well over 15 billion connected IoT devices worldwide, and the number is growing every year.

Early Milestones

An early IoT-like device was a Coca-Cola vending machine at Carnegie Mellon University in the early 1980s that students could check over the network to see whether drinks were available and cold. Radio-frequency identification (RFID) tags in the 1990s allowed tracking of goods. The arrival of cheap sensors, low-power chips, smartphones and cloud computing in the 2010s made IoT affordable and widespread. Boards such as Arduino, ESP32 and Raspberry Pi let students build their own IoT projects.

How IoT Works

An IoT system has four main layers. Sensors and devices collect data such as temperature, humidity, motion, light, location or heart rate. Connectivity sends this data using Wi-Fi, Bluetooth, Zigbee, LoRaWAN, cellular networks such as 4G and 5G, or NB-IoT. Data processing happens in the cloud or at the edge, close to the device, where software analyses the data. Finally the user interface, such as a mobile app or dashboard, shows information and allows control. Lightweight messaging protocols such as MQTT and CoAP are commonly used because IoT devices have limited power and memory. Actuators, such as motors or switches, allow the system to act, for example turning off a pump when a tank is full. Edge computing and AI let devices make decisions locally without waiting for the cloud.

IoT Devices in a Smart Home

Three IoT devices found in a smart home are smart speakers such as Amazon Echo and Google Nest, which respond to voice commands; smart thermostats and smart bulbs, which adjust temperature and lighting automatically or by phone; and smart security cameras and video doorbells, which stream video to the owner's phone. Other examples include smart door locks, smart plugs, smart TVs, robot vacuum cleaners, smart refrigerators, smoke detectors and water-tank level sensors. Wearables like fitness bands and smartwatches also count as IoT devices because they collect health data and sync it with phones.

Applications of IoT

In agriculture, soil-moisture sensors control irrigation. In healthcare, connected devices monitor patients remotely. In industry, sensors on machines predict failures, an approach called predictive maintenance, under the banner of Industry 4.0. In transport, GPS trackers manage fleets. In cities, smart streetlights and waste bins reduce energy and cost. In logistics, tags track parcels. In energy, smart meters record electricity use and send readings to the utility.

IoT in India

India uses IoT widely. FASTag, the electronic toll system that uses RFID tags on windscreens, lets vehicles pass toll plazas without stopping. Smart electricity meters are being installed across states under the Revamped Distribution Sector Scheme to reduce losses and allow remote readings. Cities under the Smart Cities Mission use connected sensors for traffic signals, water supply and waste management. Farmers use IoT-based irrigation controllers and weather stations. Indian IT firms in Chennai and Bengaluru and many startups build IoT platforms for factories and utilities.

Key Security Risk of IoT

A major security risk of IoT is that many devices are poorly protected. They often ship with weak or default passwords like "admin", rarely receive software updates, and have little computing power for strong encryption. Attackers can hack such devices to spy through cameras, steal personal data, or take control of thousands of them to form a botnet. In October 2016 the Mirai botnet, made of hacked cameras and routers, launched a huge DDoS attack that disrupted popular websites. Good practice includes changing default passwords, updating firmware, using a separate Wi-Fi network for smart devices, disabling unused features and buying from trusted brands.

Other Challenges and Future

Other challenges include privacy of collected data, lack of common standards, battery life, network reliability and e-waste. The future of IoT will be shaped by 5G, edge AI, digital twins, and low-power wide-area networks, making homes, farms, factories and cities smarter and more efficient.

Quick Facts

- IoT stands for Internet of Things: physical objects with sensors and connectivity that exchange data over the internet.
- The term was coined by Kevin Ashton in 1999.
- Smart speakers, smart bulbs or thermostats and security cameras are common smart home IoT devices.
- A key IoT security risk is weak default passwords and unpatched devices that hackers can turn into botnets.
- FASTag, based on RFID, is a widely used IoT application in India.